

Atlas Marketplace

A private-markets placement marketplace — fund managers (GPs) raise capital from verified investors (LPs). Senior Frontend Engineer task for Greenstone · Ravi Kumar

Open it

Live: atlas-marketplace-sigma.vercel.app
Code: github.com/ravikr-mca/atlas-marketplace
Sign in: click a persona card (demo only, no real accounts).
2-minute tour: **Omar** (verified investor) → open a fund → submit an indication → Dashboard. **Faisal** (pending) → every blocked action explains why. **Elena** (manager) → pipeline, decline with reason. **Layla** (admin, via mock Entra) → Approvals → approve an org, then re-open as that org.

Why this category

Mirrors Greenstone’s own business: GCC investors ↔ global fund managers. It is book-building, not e-commerce bidding: indications of interest, allocation against a hard cap, pro-rata scale-back, then subscription and funding. That shapes every technical decision below.

What is built (and runs live)

- **Discovery & diligence:** fund list (strategy filter), fund detail with Tiptap narrative, net-IRR-vs-benchmark chart, data room, live % subscribed.
- **Indication → allocation:** server-authoritative submit, **idempotency key**, scale-back at the hard cap, min-commitment, close-date and duplicate checks.
- **State machine:** 11 statuses, one transition table shared by API, UI and tests; capacity released on withdraw/decline, full book reopens.
- **Login & sessions:** persona login, staff via mock Entra SSO, 30-min sliding expiry, logout after 5 failures, deep-link return, 401 handled in one place.
- **Authorization, enforced twice:** one permissions table (role × accreditation × ownership) drives UI explanations *and* API refusals (401/403/409/422). Tenant scoping: a GP never sees another GP’s investors.
- **Trust:** verification badges, organization profiles (track record + IRR-by-vintage), **admin approvals** that change permissions immediately, audited and notified.
- **Audit trail:** every login and state change appended with actor, time and reason.
- **Seed data:** every enum value and status has data (a test fails if one doesn’t): 6 managers, 8 investors, 12 funds, 28 indications.
- **Quality:** 55 Vitest tests (allocation, state machine, permission matrix, authz, logout, expiry, approvals); responsive at 375px; reduced-motion respected; WCAG-minded contrast.

How it maps to the brief

Brief asks for	Where to see it
Buyer/seller relationship & trust	Profiles, badges, approvals, tenant-scoped views
Listing & valuation	Fund detail: terms, track record, IRR vs benchmark, data room
Bidding / purchase	Indication flow, allocation, accept → KYC → signed → funded
Concurrency & transaction state	Server-authoritative allocation, idempotency, state machine + tests
Frontend architecture	Redux = session only; TanStack Query = all server data; feature folders; typed API boundary
Security	RBAC + accreditation + ownership in UI and API; logout; audit; DEMO-labelled mock auth
Quality & delivery	Tests, CI-ready build, Vercel; written proposal + diagrams in /proposal

Decisions worth discussing

- **Redux and TanStack Query?** Different jobs: Redux never caches API data; the identity (“me”) is a query, so approvals apply with no reload.
- **Mock API, not MSW:** functions return copies across a simulated network boundary; MSW is the first thing I’d add.
- **Tokens in sessionStorage** is a prototype shortcut; production = httpOnly Secure SameSite cookie (Sanctum) + Entra ID.
- **Visual:** Greenstone green foundation; sky-blue reserved for investor actions.

Honest scope — designed in the proposal, not yet in the prototype

Messaging threads · listing-creation wizard · GP allocation override · full discovery filters (geography, vintage, ticket fit) · glossary/consent gates (NDA, risk acknowledgement) · route-level code splitting (bundle ≈ 2 MB) · real-time (Pusher) · Playwright E2E. Each is specified in proposal/technical-approach.md with when and why I’d add it; I chose depth in the transaction and trust core over breadth.

AI disclosure: I directed every decision; Claude Code executed. Demo credentials are mock, test-only and reserved-domain; no real people or data.